

3.6 More Operations on Sets

Indexed Families of Sets (The Union and Intersection of a Family of Sets)

Definition. Let $\mathcal{C}$ be a family of sets. The **union** of $\mathcal{C}$ is defined as the set of all elements that are in at least one of the sets in $\mathcal{C}$. We write

$$\bigcup_{X \in \mathcal{C}} X = \{x \in E \mid x \in X \text{ for some } X \in \mathcal{C}\}, \text{ where } U \text{ is the universal set.}$$

The **intersection** of $\mathcal{C}$ is defined as the set of all elements that are in all of the sets in $\mathcal{C}$. That is,

$$\bigcap_{X \in \mathcal{C}} X = \{x \in E \mid x \in X \text{ for all } X \in \mathcal{C}\}$$

For example, let the universal set is $\mathbb{N}$ and

$A = \{1, 2, 3, 4, 5\}$, $B = \{2, 3, 4, 5, 6\}$, $C = \{3, 4, 5, 6, 7\}$, $D = \{4, 5, 6, 7, 8\}$,
 $S = \{5, 6, 7, 8, 9\}$ and $T = \{6, 7, 8, 9, 10\}$, where $E = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13\}$.

We can then consider the following families of sets:

$$\mathcal{A} = \{A, B, C, D\} \text{ and } \mathcal{B} = \{A, B, C, D, S, T\}.$$

Explain why

$$\bigcup_{X \in \mathcal{A}} X = A \cup B \cup C \cup D \quad \text{and} \quad \bigcap_{X \in \mathcal{A}} X = A \cap B \cap C \cap D,$$

An Indexed Family of Sets

We often use subscripts to identify sets. For example, instead of using A, B, C , and D as the names of the sets, we could have used A_1, A_2, A_3 , and A_4 .

When we do this, we are using the subscript as an identifying tag, or index, for each set. We can also use this idea to specify an infinite family of sets.

For example, for each natural number n , we define

$$C_n = \{n, n+1, n+2, n+3, n+4\}.$$

$$\begin{aligned} C_1 &= \{1, 2, 3, 4, 5\}, C_2 = \{2, 3, 4, 5, 6\} \\ C_3 &= \{3, 4, 5, 6, 7\}, \dots, C_n, \dots \\ \mathcal{C} &= \{C_1, C_2, C_3, \dots, C_n, \dots\} \\ &= \{C_n \mid n \in \mathbb{N}\} \end{aligned}$$

The Union and Intersection of an Indexed Family of Sets

Let n be a natural number and $\mathcal{A} = \{A_1, A_2, \dots, A_n\}$ is a family of n sets, then the union of these n sets, denoted by $A_1 \cup A_2 \cup \dots \cup A_n$ or $\bigcup_{j=1}^n A_j$, is defined as

$$\bigcup_{j=1}^n A_j = \{x \in U \mid x \in A_j, \text{ for some } j \text{ with } 1 \leq j \leq n\}.$$

We can also define the intersection of these n sets, denoted by $A_1 \cap A_2 \cap \dots \cap A_n$ or $\bigcap_{j=1}^n A_j$, as

$$\bigcap_{j=1}^n A_j = \{x \in U \mid x \in A_j, \text{ for all } j \text{ with } 1 \leq j \leq n\}.$$

We can also extend this idea to define the union and intersection of a family that consists of infinitely many sets. So if $\mathcal{B} = \{B_1, B_2, \dots, B_n, \dots\}$, then

$$\bigcup_{j=1}^{\infty} B_j = \{x \in U \mid x \in B_j, \text{ for some } j \text{ with } j \geq 1\}, \text{ and}$$

$$\bigcap_{j=1}^{\infty} B_j = \{x \in U \mid x \in B_j, \text{ for all } j \text{ with } j \geq 1\}.$$

For each natural number n , let $A_n = \{1, n, n^2\}$. For example,

$$A_1 = \{1\}$$

$$A_2 = \{1, 2, 4\},$$

$$A_3 = \{1, 3, 9\},$$

and

$$\bigcup_{j=1}^3 A_j = \{1, 2, 3, 4, 9\},$$

$$\bigcap_{j=1}^3 A_j = \{1\}.$$

Determine each of the following sets:

* 1. $\bigcup_{j=1}^6 A_j$

* 3. $\bigcup_{j=3}^6 A_j$

* 5. $\bigcup_{j=1}^{\infty} A_j$

* 2. $\bigcap_{j=1}^6 A_j$

4. $\bigcap_{j=3}^6 A_j$

* 6. $\bigcap_{j=1}^{\infty} A_j = \{1\}$

In all of the examples we have studied so far, we have used $\mathbb{N}$ or a subset of $\mathbb{N}$ to index or label the sets in a family of sets. We can use other sets to index or label sets in a family of sets. For example, for each real number x , we can define B_x to be the closed interval $[x, x + 2]$. That is,

$$B_x = \{y \in \mathbb{R} \mid x \leq y \leq x + 2\}.$$

So we make the following definition. In this definition, Λ is the uppercase Greek letter lambda and α is the lowercase Greek letter alpha.

Definition. Let Λ be a nonempty set and suppose that for each $\alpha \in \Lambda$, there is a corresponding set A_α . The family of sets $\{A_\alpha \mid \alpha \in \Lambda\}$ is called an **indexed family of sets** indexed by Λ . Each $\alpha \in \Lambda$ is called an **index** and Λ is called an **indexing set**.

Definition. Let Λ be a nonempty indexing set and let $\mathcal{A} = \{A_\alpha \mid \alpha \in \Lambda\}$ be an indexed family of sets. The **union over** $\mathcal{A}$ is defined as the set of all elements that are in at least one of sets A_α , where $\alpha \in \Lambda$. We write

$$\bigcup_{\alpha \in \Lambda} A_\alpha = \{x \in U \mid \text{there exists an } \alpha \in \Lambda \text{ with } x \in A_\alpha\}.$$

The **intersection over** $\mathcal{A}$ is the set of all elements that are in all of the sets A_α for each $\alpha \in \Lambda$. That is,

$$\bigcap_{\alpha \in \Lambda} A_\alpha = \{x \in U \mid \text{for all } \alpha \in \Lambda, x \in A_\alpha\}.$$

$$a_1, a_2, a_3, \dots, a_n$$

$$a_1 + a_2 + a_3 + \dots + a_n = \sum_{i=1}^n a_i$$

$$a_1 + a_2 + a_3 + \dots = \sum_{n=1}^{\infty} a_n$$

Note: i) Let x be an arbitrary element

Suppose $x \in \bigcup_{j=1}^{\infty} B_j$

Then, $x \in B_{j_0}$ for some $j_0 \geq 1$

ii) Suppose $x \in \bigcap_{j=1}^{\infty} B_j$

Then, $x \in B_j$ for all $j \geq 1$

Ex Prove that $\forall n \in \mathbb{N}, \bigcap_{j=1}^n A_j = \{1\}$
(use M.I.)

Eg

$$\Lambda = \{\alpha \in \mathbb{R} \mid -1 < \alpha < 1\}$$

$$\Lambda = \{\alpha \in \mathbb{R} \mid 0 < \alpha < 3\}$$

Note: i) Let x be an arbitrary element

Suppose $x \in \bigcup_{\alpha \in \Lambda} A_\alpha$

Then, $x \in U$ and $x \in A_{\alpha_0}$ for some $\alpha_0 \in \Lambda$

ii) Suppose $x \in \bigcap_{\alpha \in \Lambda} A_\alpha$

Then $x \in U$ and $x \in A_\alpha$ for all $\alpha \in \Lambda$

Theorem 01. Let Λ be a nonempty indexing set and let $\mathcal{A} = \{A_\alpha \mid \alpha \in \Lambda\}$ be an indexed family of sets, each of which is a subset of some universal set U . Then

$$1. \text{ For each } \beta \in \Lambda, \bigcap_{\alpha \in \Lambda} A_\alpha \subseteq A_\beta.$$

$$2. \text{ For each } \beta \in \Lambda, A_\beta \subseteq \bigcup_{\alpha \in \Lambda} A_\alpha.$$

$$3. \left(\bigcap_{\alpha \in \Lambda} A_\alpha \right)^c = \bigcup_{\alpha \in \Lambda} A_\alpha^c$$

$$4. \left(\bigcup_{\alpha \in \Lambda} A_\alpha \right)^c = \bigcap_{\alpha \in \Lambda} A_\alpha^c$$

Ex.

Parts (3) and (4) are known as **De Morgan's Laws**.

Theorem 02. Let Λ be a nonempty indexing set, let $\mathcal{A} = \{A_\alpha \mid \alpha \in \Lambda\}$ be an indexed family of sets, and let B be a set. Then

$$1. B \cap \left(\bigcup_{\alpha \in \Lambda} A_\alpha \right) = \bigcup_{\alpha \in \Lambda} (B \cap A_\alpha), \text{ and}$$

Ex.

$$2. B \cup \left(\bigcap_{\alpha \in \Lambda} A_\alpha \right) = \bigcap_{\alpha \in \Lambda} (B \cup A_\alpha).$$

End of Dis

Pairwise Disjoint Families of Sets

Two sets A and B to be disjoint provided that $A \cap B = \emptyset$.

In a similar manner, if Λ is a nonempty indexing set and $\mathcal{A} = \{A_\alpha \mid \alpha \in \Lambda\}$ is an indexed family of sets, we can say that this indexed family of sets is **disjoint** provided that $\bigcap_{\alpha \in \Lambda} A_\alpha = \emptyset$. However, we can use the concept of two disjoint sets to define a somewhat more interesting type of "disjointness" for an indexed family of sets.

Definition. Let Λ be a nonempty indexing set, and let $\mathcal{A} = \{A_\alpha \mid \alpha \in \Lambda\}$ be an indexed family of sets. We say that $\mathcal{A}$ is **pairwise disjoint** provided that for all α and β in Λ , if $A_\alpha \neq A_\beta$, then $A_\alpha \cap A_\beta = \emptyset$.

Note:

The **Cartesian product** (or just **product**) of two sets A and B is the set

$$A \times B := \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

The product of a finite family $A_1, A_2, \dots, A_n$ of sets is the set

$$A_1 \times A_2 \times \cdots \times A_n := \{(a_1, a_2, \dots, a_n) \mid a_1 \in A_1, a_2 \in A_2, \dots, a_n \in A_n\}.$$

The product of a finite family $A_1, A_2, \dots, A_n$ of sets is the set

$$\begin{aligned} \prod_{i=1}^n A_i &= A_1 \times A_2 \times \cdots \times A_n \\ &= : \{(a_1, a_2, \dots, a_n) \mid a_1 \in A_1, a_2 \in A_2, \dots, a_n \in A_n\}. \end{aligned}$$

Ex.

01. For each natural number n , let $A_n = \{x \in \mathbb{R} \mid n-1 < x < n\}$. Prove that $\{A_n \mid n \in \mathbb{N}\}$ is a pairwise disjoint family of sets and that $\bigcup_{n \in \mathbb{N}} A_n = (\mathbb{R}^+ - \mathbb{N})$.

02. For each natural number n , let $A_n = \{k \in \mathbb{N} \mid k \geq n\}$. Determine if the following statements are true or false. Justify each conclusion.

(a) For all $j, k \in \mathbb{N}$, if $j \neq k$, then $A_j \cap A_k \neq \emptyset$.

(b) $\bigcap_{k \in \mathbb{N}} A_k = \emptyset$

03. Give an example of an indexed family of sets $\{A_n \mid n \in \mathbb{N}\}$ such all three of the following conditions are true:

(i) For each $m \in \mathbb{N}$, $A_m \subseteq (0, 1)$;

(ii) For each $j, k \in \mathbb{N}$, if $j \neq k$, then $A_j \cap A_k \neq \emptyset$; and

(iii) $\bigcap_{k \in \mathbb{N}} A_k = \emptyset$.

04. Prove the following:

(a) Let $\{A_i : i \in I\}$ and $\{B_i : i \in I\}$ be indexed families of sets with the same indexed set I . Suppose $A_i \subseteq B_i$ for all $i \in I$.

(i) $\bigcup_{i \in I} A_i \subseteq \bigcup_{i \in I} B_i$.

(ii) $\bigcap_{i \in I} A_i \subseteq \bigcap_{i \in I} B_i$.

(b) Let $\{A_i : i \in I\}$ and $\{B_j : j \in J\}$ be indexed families of sets.

If there is an $i_0 \in I$ such that $A_{i_0} \subseteq B_j$ for all $j \in J$, then $\bigcap_{i \in I} A_i \subseteq \bigcap_{j \in J} B_j$.

(c) Suppose that A is a set and that $\{B_i : i \in I\}$ is an indexed family of sets. Then $A \cap \bigcup_{i \in I} B_i = \bigcup_{i \in I} (A \cap B_i)$.

(d) Suppose that A is a set and that $\{B_i : i \in I\}$ is an indexed family of sets.

Then $A \cup \bigcap_{i \in I} B_i = \bigcap_{i \in I} (A \cup B_i)$.

(f) Suppose that A is a set and that $\{B_i : i \in I\}$ is an indexed family of sets.

Then $A \setminus \bigcap_{i \in I} B_i = \bigcup_{i \in I} (A \setminus B_i)$.

4 Relations